CAPSTONE 12 — DBMS and SQL Practice Companion

Research report and implementation summary

Project question: How can AI help students learn database design and SQL through guided practice and error explanation?

Target audience

Third-semester Computer Science and Engineering students with introductory programming knowledge and limited hands-on SQL experience.

Objectives

Learn ER modelling, map entities to relational tables, explain 1NF–3NF, practise SELECT queries, joins, grouping, subqueries, indexing and transaction concepts.

AI research tools

ChatGPT and Google Gemini can be used as two AI research assistants to draft explanations, candidate SQL exercises and quiz questions. Compare their outputs, then verify claims against official database documentation and execute SQL. AI output is a draft, not an authority.

Database design

The fictional library schema contains Authors, Books, Members, Loans and LoanItems. Relationships: one author to many books; one member to many loans; one loan to many loan items; one book may appear in many loan items. LoanItems uses a composite primary key (loan_id, book_id).

Normalization

UNF stores repeating book groups in a loan record. 1NF makes values atomic and each row represent one loaned book. 2NF separates member and book attributes that depend on only part of the composite key. 3NF separates author details into Authors so author_name depends on author_id, not transitively on book_id.

Application

A local Flask application uses SQLite. It provides a read-only SELECT console, a 1NF-to-3NF guide, a 12-question quiz and a rule-based SQLite error explanation helper. The browser blocks write/DDL commands and multiple statements.

Verification

All 25 supplied practice statements were executed successfully against the seeded database. Smoke tests: 8 books, 5 members, two Romance books, two Fiction books and one overdue loan.

Ethics and limitations

All records are fictional and use example.test email addresses. No real student or patient data is used. Error explanations are heuristic and may be incomplete; students must inspect the schema and test proposed fixes. This is an educational prototype, not production software.

References

SQLite documentation: https://www.sqlite.org/docs.html; SELECT: https://www.sqlite.org/lang_select.html; query planner: https://www.sqlite.org/eqp.html; PostgreSQL: https://www.postgresql.org/docs/current/; MySQL: https://dev.mysql.com/doc/refman/8.4/en/; Flask: https://flask.palletsprojects.com/.